

Worksheet — Motion in a Plane

Total marks: 29 · Questions: 12

Instructions: Attempt ALL questions. Each question is followed by its mark allocation. Take $g = 10 \text{ m/s}^2$ unless otherwise stated. Show working for subjective and case-based questions. The total marks for this worksheet is 29.

Q1. Which of the following is a vector quantity? [1 mark]

- (A) mass
- (B) energy
- (C) displacement
- (D) temperature

Q2. Two perpendicular forces of magnitudes 6 N and 8 N act on a body. The magnitude of the resultant is [1 mark]

- (A) 2 N
- (B) 10 N
- (C) 14 N
- (D) 48 N

Q3. On level ground, the range of a projectile is maximum when the launch angle is [1 mark]

- (A) 30°
- (B) 45°
- (C) 60°
- (D) 90°

Q4. A particle moves in a circle of radius 2 m at speed 4 m/s. Its centripetal acceleration is [1 mark]

- (A) 2 m/s^2
- (B) 4 m/s^2
- (C) 8 m/s^2
- (D) 16 m/s^2

Q5. The horizontal velocity of a projectile (no air resistance) stays constant throughout its flight. [1 mark]

- (A) True
- (B) False

Q6. Centripetal force is a new kind of fundamental force, separate from gravity, tension and friction. [1 mark]

(A) True

(B) False

Q7. The maximum height of a projectile launched at speed u and angle θ is $H = \underline{\hspace{2cm}}$. [1 mark]

Q8. The velocity of A relative to B is given by $v_{AB} = \underline{\hspace{2cm}}$. [1 mark]

Q9. A projectile is launched at 30 m/s at 30° above the horizontal. Find its time of flight, maximum height, and range. ($g = 10 \text{ m/s}^2$) [3 marks]

Q10. Explain why uniform circular motion is accelerated motion even though the speed is constant. [3 marks]

Q11. Derive expressions for the time of flight, maximum height and horizontal range of a projectile launched at speed u and angle θ above the horizontal on level ground. [5 marks]

Q12. Read the passage and answer the questions that follow. A car of mass 1000 kg negotiates a flat (un-banked) circular curve of radius 50 m at a speed of 15 m/s. The coefficient of friction between the tyres and the road is 0.5. Take $g = 10 \text{ m/s}^2$. (i) What centripetal force is required for the car to make this turn? (ii) What is the maximum friction force the road can supply? (iii) Can the car make the turn safely? Explain. (iv) What is the maximum safe speed on this curve? [10 marks]

Answer Key

Worksheet — Motion in a Plane

Q1. Which of the following is a vector quantity?

Answer: displacement

Displacement has both magnitude and a direction, so it is a vector. The others are scalars.

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Q2. Two perpendicular forces of magnitudes 6 N and 8 N act on a body. The magnitude of the resultant is

Answer: 10 N

$$R = \sqrt{6^2 + 8^2} = \sqrt{100} = 10 \text{ N.}$$

MCQ · EASY · 12-NIOS-PHY-MIP-03

Q3. On level ground, the range of a projectile is maximum when the launch angle is

Answer: 45°

$$R = u^2 \sin 2\theta / g; \sin 2\theta \text{ is maximum at } 2\theta = 90^\circ, \text{ i.e. } \theta = 45^\circ.$$

MCQ · EASY · 12-NIOS-PHY-MIP-07

Q4. A particle moves in a circle of radius 2 m at speed 4 m/s. Its centripetal acceleration is

Answer: 8 m/s²

$$a_c = v^2 / r = 16 / 2 = 8 \text{ m/s}^2.$$

MCQ · EASY · 12-NIOS-PHY-MIP-08

Q5. The horizontal velocity of a projectile (no air resistance) stays constant throughout its flight.

Answer: True

No horizontal force acts, so $a_x = 0$ and v_x stays at $u \cos \theta$ throughout.

TRUE_FALSE · EASY · 12-NIOS-PHY-MIP-06

Q6. Centripetal force is a new kind of fundamental force, separate from gravity, tension and friction.

Answer: False

Centripetal force is the NET inward force; its physical origin is one of the existing forces (tension, gravity, friction, normal etc.).

TRUE_FALSE · EASY · 12-NIOS-PHY-MIP-09

Q7. The maximum height of a projectile launched at speed u and angle θ is $H = \underline{\hspace{2cm}}$.

Answer: $u^2 \sin^2 \theta / (2g)$

$$H = u_y^2 / (2g) = u^2 \sin^2 \theta / (2g).$$

FILL_BLANK · EASY · 12-NIOS-PHY-MIP-06

Q8. The velocity of A relative to B is given by $v_{AB} = \underline{\hspace{2cm}}$.

Answer: $v_A - v_B$

Relative velocity is a vector subtraction.

FILL_BLANK · EASY · 12-NIOS-PHY-MIP-10

Q9. A projectile is launched at 30 m/s at 30° above the horizontal. Find its time of flight, maximum height, and range. ($g = 10 \text{ m/s}^2$)

Answer: $u_x = 30 \cos 30^\circ = 15\sqrt{3} \text{ m/s}$; $u_y = 30 \sin 30^\circ = 15 \text{ m/s}$. $T = 2u \sin \theta / g = 30/10 = 3 \text{ s}$. $H = u^2 \sin^2 \theta / (2g) = 900 \times 0.25 / 20 = 11.25 \text{ m}$. $R = u^2 \sin 2\theta / g = 900 \times \sin 60^\circ / 10 = 45\sqrt{3} \approx 77.9 \text{ m}$.

Apply the standard projectile formulas T, H, R component by component.

SHORT_ANSWER · MEDIUM · 12-NIOS-PHY-MIP-06

Q10. Explain why uniform circular motion is accelerated motion even though the speed is constant.

Answer: Acceleration is the rate of change of velocity (a vector), not speed (a scalar). In uniform circular motion the direction of velocity changes continuously, so the velocity vector itself changes — there is a non-zero acceleration (the centripetal acceleration, v^2/r , directed toward the centre).

The key distinction is between speed (scalar) and velocity (vector).

SHORT_ANSWER · MEDIUM · 12-NIOS-PHY-MIP-08

Q11. Derive expressions for the time of flight, maximum height and horizontal range of a projectile launched at speed u and angle θ above the horizontal on level ground.

Answer: Take launch point as origin, x horizontal in direction of motion, y vertical upward. $u_x = u \cos \theta$ (constant), $u_y = u \sin \theta$ (decreasing under gravity). $a_x = 0$, $a_y = -g$. So $x = u \cos \theta \cdot t$ and $y = u \sin \theta \cdot t - (1/2) g t^2$. Time of flight T: set $y = 0$ (projectile returns to launch height): $u \sin \theta \cdot t - (1/2) g t^2 = 0 \Rightarrow t = 0$ or $t = 2 u \sin \theta / g$. The non-trivial root is $T = 2 u \sin \theta / g$. Maximum height H: $v_y = 0$ at apex. Use $v_y^2 = u_y^2 - 2 g H \Rightarrow H = u^2 \sin^2 \theta / (2g)$. Range R: $R = u_x \times T = u \cos \theta \times 2 u \sin \theta / g = u^2 \sin 2\theta / g$. Notes: range is maximised at $\theta = 45^\circ$ (where $\sin 2\theta = 1$); complementary angles θ and $(90^\circ - \theta)$ give the same range.

Standard derivation using component-wise 1D kinematics.

LONG_ANSWER · MEDIUM · 12-NIOS-PHY-MIP-06

Q12. Read the passage and answer the questions that follow. A car of mass 1000 kg negotiates a flat (un-banked) circular curve of radius 50 m at a speed of 15 m/s. The coefficient of friction between the tyres and the road is 0.5. Take $g = 10 \text{ m/s}^2$. (i) What centripetal force is required for the car to make this turn? (ii) What is the maximum friction force the road can supply? (iii) Can the car make the turn safely? Explain. (iv) What is the maximum safe speed on this curve?

Answer: (i) $F_c = m v^2 / r = 1000 \times 225 / 50 = 4500 \text{ N}$, directed toward the centre. (ii) $f_{\text{max}} = \mu m g = 0.5 \times 1000 \times 10 = 5000 \text{ N}$. (iii) Yes — required force (4500 N) is less than available friction (5000 N), so the tyres can supply the inward force without skidding. (iv) $v_{\text{max}} = \sqrt{\mu g r} = \sqrt{(0.5 \times 10 \times 50)} = \sqrt{250} \approx 15.8 \text{ m/s}$. The driver is at 15 m/s, just below this safe limit.

Case-based questions test whether students can apply the centripetal-force concept to a real situation while keeping track of which physical agent supplies the force.

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